

Final Exam Review Lab: Part 2

IMPORTANT INFO Large Language Model usage is ALLOWED for this assignment, but MUST be conducted through Google Gemini using your NCSU account (<https://oit.ncsu.edu/2024/11/20/gemini-chatbot-available-to-nc-state/>).

IF YOU USE GEMINI TO COMPLETE THIS LAB, you are required to submit your chat logs. Please follow the Lab3 Gemini Log Extraction guide for details on how to do so.

IF YOU USE GEMINI TO COMPLETE A QUESTION ON THIS LAB, please fill in the corresponding white square (☐) next to the question text.

Your submission should consist of this PDF with your answers filled in, and any additional pages of work you may need to complete each problem.

Note This lab continues the Final Exam Review from Part 1 (Questions 1–3). The questions below are numbered 4 through 12.

Question 4:

Part 1 You are still the restaurant manager, and now you are stocking the dessert tray. The tray has 8 positions referred to by the universe $U = \{1, 2, \dots, 8\}$. Positions $A = \{1, 2, 3, 4\}$ must hold cakes, and positions $B = \{3, 4, 5, 6\}$ must hold items containing chocolate.

Compute each of the following:

1. $A \cup B$ ☐

2. $A \cap B$ ☐

3. $A - B$ ☐

4. $A \oplus B$ ☐

5. $\overline{A}$ ☐

Part 2 Compute $|A \times B|$ and $|\mathcal{P}(A)|$. For each, explain what the quantity counts.
☐

Part 3 For each function below, determine whether it is injective, surjective, bijective, or none of these.

1. $f_1 : \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$ where $f_1(1) = a$, $f_1(2) = b$, $f_1(3) = c$, $f_1(4) = a$ $\square$

2. $f_2 : \{1, 2, 3\} \rightarrow \{a, b, c, d\}$ where $f_2(1) = b$, $f_2(2) = d$, $f_2(3) = a$ $\square$

3. $f_3 : \{1, 2, 3\} \rightarrow \{a, b, c\}$ where $f_3(1) = c$, $f_3(2) = a$, $f_3(3) = b$ $\square$

Part 4 A sundae comes with your choice of the 8 available toppings. How many different sundaes can you make, assuming each topping is either on the sundae or not? Exhibit the bijection to binary strings of length 8 that justifies your count, and cite the counting concepts you use. $\square$

Part 5 How many strings are there over the set $\{a, b, c\}$ that have length 5 and contain at least one 'a'? Count by complement.

□

Part 6 How many strings are there over the set $\{a, b, c\}$ that have length 5 and start with 'a' or end with 'b'? Cite the counting concept you use. □

Part 7 You place 25 desserts into the 7 display cases at the front of the restaurant. What is the minimum number of desserts the case with the maximum number of desserts across the 7 display cases can hold? □

Question 5:

Recall the salad bar from Question 1, where $O_{x,y}$ represents an open slot for an item to be placed in:

$$\begin{array}{cccc} O_{00} & O_{01} & O_{02} & O_{03} \\ O_{10} & O_{11} & O_{12} & O_{13} \\ & O_{21} & O_{22} & \end{array}$$

Suppose you assign the items i_0 through i_9 to the ten open slots at random, with every assignment equally likely.

Define the Top and Bottom rows of the salad bar:

$$\forall O_{x,y} : [x = 0] \leftrightarrow [O_{x,y} \in \text{Top}]$$

$$\forall O_{x,y} : [x = 2] \leftrightarrow [O_{x,y} \in \text{Bottom}]$$

Define the events:

- T_9 : i_9 is placed in a slot in Top
- T_8 : i_8 is placed in a slot in Top

Part 1 What is $p(T_9)$? $\square$

Part 2 What is $p(T_9 \cap T_8)$? $\square$

Part 3 Find the probability that i_9 is placed in a slot in Top or a slot in Bottom. $\square$

Part 4 Find $p(T_8 \cup T_9)$. $\square$

Part 5 Find the probability that i_0 is *not* placed in slot O_{00} . $\square$

Part 6 What is $p(i_9 \text{ placed in } O_{00} \mid T_9)$? $\square$

Part 7 Are the events T_8 and T_9 independent? $\square$

Part 8 Prep station A prepares 60% of the items and mislabels 2% of what it prepares. Prep station B prepares 40% of the items and mislabels 5% of what it prepares. An item is selected at random. If the chosen item is mislabeled, what is the probability that it came from station B? $\square$

Question 6:

Your kitchen has a sensor network. Define the propositions:

- p : the oven is on
- q : the door is open
- r : the alarm sounds

Part 1 Translate into a propositional expression: “If the oven is on and the door is open, then the alarm sounds.” $\square$

Part 2 Translate into natural English: $\neg r \rightarrow (\neg p \vee \neg q)$ $\square$

Part 3 Use a truth table to determine whether $(p \wedge q) \rightarrow r$ is satisfiable, a tautology, or a contradiction. If it is satisfiable but not a tautology, indicate one row where it is 1 and one row where it is 0.

p	q	r	$A : p \wedge q$	$A \rightarrow r$
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

$\square$

Part 4 Simplify $\neg(p \wedge \neg q) \vee q$ to a minimal equivalent expression. Cite the numbered law from the logical equivalences sheet for EVERY step. $\square$

Question 7:

Part 1 Prove that $\forall x, y \in \mathbb{Z}$, if x and y are even, then $x + y$ is even. $\square$

Part 2 Prove that $\forall n \in \mathbb{Z}$, $n^2 \equiv 0 \pmod{4}$ or $n^2 \equiv 1 \pmod{4}$. (Consider two cases.) $\square$

Part 3 Prove that $\forall n \in \mathbb{Z}$, if n^2 is even, then n is even. State the contrapositive explicitly before proving it. $\square$

Part 4 Prove by contradiction that the sum of a rational number and an irrational number is irrational. $\square$

Part 5 Prove that $\forall a, b, c, d \in \mathbb{Z}$, if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$. $\square$

Part 6 Prove that the following statement is **false**:
For all $a, b \in \mathbb{Z}$, if $a^2 \equiv b^2 \pmod{4}$, then $a \equiv b \pmod{4}$. $\square$

Question 8:

Part 1 Consider the sequence 4, 7, 10, 13, 16, ... Give (a) a recursive form with its base case, and (b) a closed form. $\square$

Part 2 For each recurrence below: is it an LHRR? Give one sentence for each property of an LHRR.

1. $a_n = 2a_{n-1} + 3$ $\square$

2. $a_n = a_{n-1} \cdot a_{n-2}$ $\square$

3. $a_n = 5a_{n-1} - 6a_{n-2}$ $\square$

Part 3 Recall the Squemp Numbers from Lab 3 Part 1 Question 2. Enumerating them from $q_0 = 0$ by the successor rule gives the recursive form:

$$\begin{aligned}q_n &= 2q_{n-1} + 3 \\ q_0 &= 0\end{aligned}$$

Prove by induction that the closed form $q_n = 3(2^n - 1)$ satisfies this recursive form for all $n \geq 0$. $\square$

Part 4 Find closed form of this first order LHR:

$$\begin{aligned}S_n &= 3 \cdot S_{n-1} \\ S_0 &= 2\end{aligned}$$

Verify your closed form against S_1 and S_2 . $\square$

Part 5 Find closed form of this second order LHR. Show the characteristic equation, its roots, and how you solve for the constants.

$$a_n = 5a_{n-1} - 6a_{n-2}$$

$$a_0 = 1$$

$$a_1 = 4$$

□

Question 9:

Part 1 Lemma 1: $\phi(p) = p - 1$ for all primes p .

Lemma 2: For any primes p, q , if $p \neq q$, then $\phi(p \cdot q) = \phi(p) \cdot \phi(q)$

Compute $\phi(15)$, and explain which method you used ((a) direct counting, (b) Lemma 1 for primes, or (c) Lemma 2 for products of distinct primes). Then state the value of $\phi(p)$ for any prime p , with one sentence of justification. $\square$

Part 2 Recall that an encryption scheme is composed of 3 (polynomial time) functions: $\text{KeyGen}(n)$, $\text{Enc}(m, k)$, and $\text{Dec}(c, k)$. Define all three functions for the Caesar Cipher. $\square$

Part 3 Theorem (Euler's): Given $A, n, x \in \mathbb{Z}_{>0}$ such that A and n are coprime,

$$A^{\phi(n) \cdot x} \bmod n = 1$$

Recall the Proposition from the cryptography lecture: if p, q are distinct primes and $e, d \in \mathbb{Z}_{>0}$ are such that

$$1 < e, d < (p-1)(q-1) \text{ and } ed \equiv 1 \pmod{(p-1)(q-1)},$$

then for every $m \in \mathbb{Z}_{>0}$ we have $m^{ed} \equiv m \pmod{pq}$.

Let $N = p \cdot q$. Prove $m^{ed} \equiv m \pmod{pq}$ for the case where m and N are coprime.
 $\square$

Question 10:

Part 1 Order these functions from slowest growing to fastest growing, and give the tightest Θ class for each:

1. $5n^2 + 3n$

2. $n \log n + 20n$

3. $2^n + n^3$

4. 100

5. $n! + 2^n$

□

Part 2 Prove

$$5n^2 + 3n = O(n^2)$$

Show how you get the constants c and k , and verify the inequality. □

Part 3 Is $5n^2 + 3n$ also $\Theta(n^2)$ Provide the c_1 , c_2 , and k you will use to verify the relation.

Part 4 Recall the salad bars from Question 1 as S_1 . We want to build a new salad bar with an identical layout of slots. The following algorithm builds S_2 from S_1 by shifting every item one slot to the right within its row, wrapping to the first slot of the same row:

```
ShiftBar(S1):  
    for each row r in S1:                # rows have widths 4, 4, 2  
        w = width(r)  
        for j = 0 to w - 1:  
            S2[r][(j + 1) mod w] = S1[r][j]  
    return S2
```

Explain the run-time of **ShiftBar**. Let n be the total number of slots; state the time complexity in Θ notation. $\square$

Question 11:

Part 1 Recall the salad bar items i_0 through i_9 from Question 1. Propose your own binary relation R on the set $\{i_0, \dots, i_9\}$. Give the rule for your relation, enumerate its pairs, and draw its arrow diagram. Then determine whether *your* relation is reflexive, irreflexive, symmetric, anti-symmetric, asymmetric, and transitive. Justify each answer with a witness or a short argument. $\square$

Part 2 Your six chefs trained each other over the years: p_0 trained p_1 and p_2 ; p_1 trained p_3 and p_4 ; p_2 trained p_5 .

Define the relationship S on $P = \{p_0, \dots, p_5\}$: $x S y$ if x is a trainee of y ; specifically, x trained under y directly or through a chain of teachers. Each chef counts as a student of themselves.

1. Enumerate the relationship S and draw its arrow diagram. $\square$

2. Determine whether S is reflexive, irreflexive, symmetric, anti-symmetric, asymmetric, and transitive. Justify each answer. $\square$

3. Is S is a partial order? If so, draw its Hasse diagram. $\square$

4. If appropriate, identify all maximal and all minimal elements of (P, S) . $\square$

Part 3 Define the relationship T on $B = \{0, 1, \dots, 8\}$: $a T b$ if $a \pmod{2} \equiv b \pmod{3}$. Determine whether S is reflexive, irreflexive, symmetric, anti-symmetric, asymmetric, and transitive. Is it a poset? Equivalence relation? $\square$

Question 12:

Your restaurant has grown into six locations connected by delivery routes.

Part 1 Draw the undirected graph $G = (V, E)$ where $V = \{A, B, C, D, E, F\}$ and

$$E = \{\{A, B\}, \{A, C\}, \{B, C\}, \{B, D\}, \{C, D\}, \{D, E\}, \{D, F\}, \{E, F\}\}$$

Give the degree of each vertex. $\square$

Part 2 Classify each sequence as a walk, trail, path, circuit, and/or cycle. List every label that applies, with one sentence for each.

1. A, B, C, A $\square$

2. A, B, D, C, B $\square$

3. A, C, D, E, F $\square$

Part 3 Does G have an Euler circuit? An Euler trail? Justify your answers using the vertex degrees, and exhibit an Euler trail if one exists. $\square$

Part 4 Exhibit a Hamilton path in G . Does G have a Hamilton cycle? Explain your reasoning. $\square$

Part 5 Draw a spanning tree of G . How many edges must any spanning tree of G have? $\square$

Part 6 Perform Breadth-First-Search on G , starting from node 'A' and visiting neighbors in alphabetical order. Then perform Depth-First-Search the same way. Give the visitation order for each, and show your trace. $\square$